SHETH L.U.J AND SIR M.V. COLLEGE

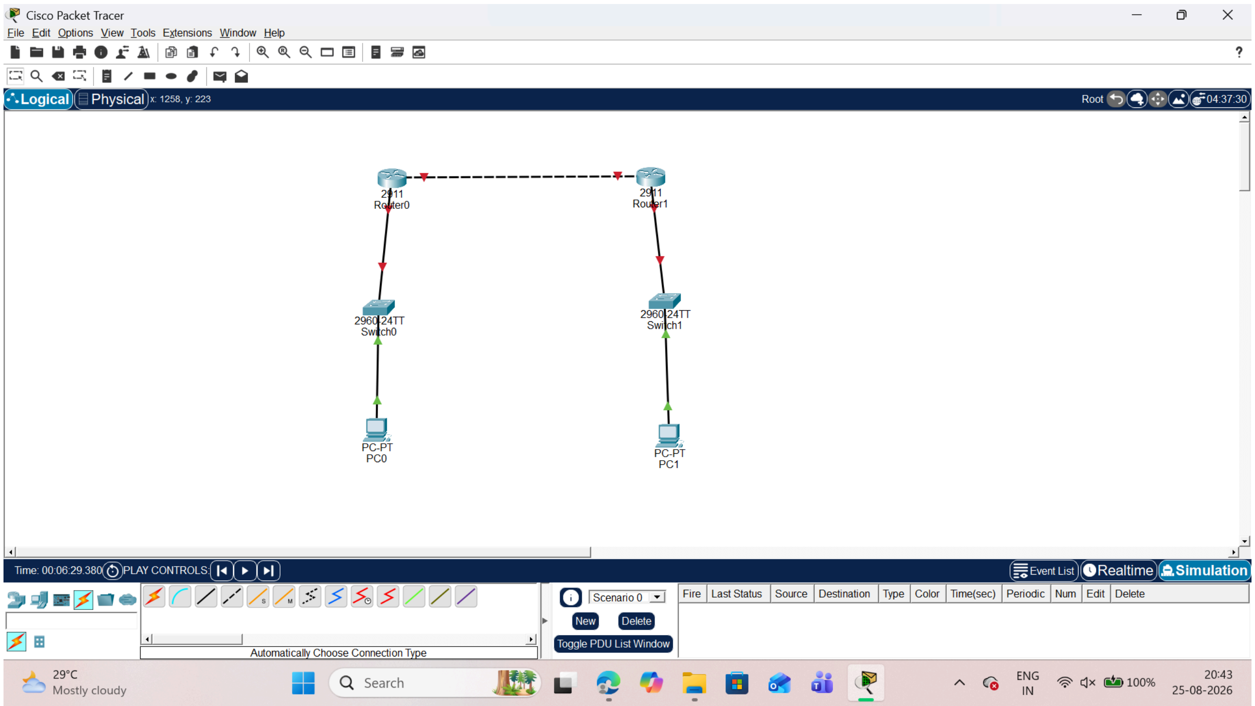
SUBJECT: CYBER & INFORMATION SECURITY

PRACTICAL - 6

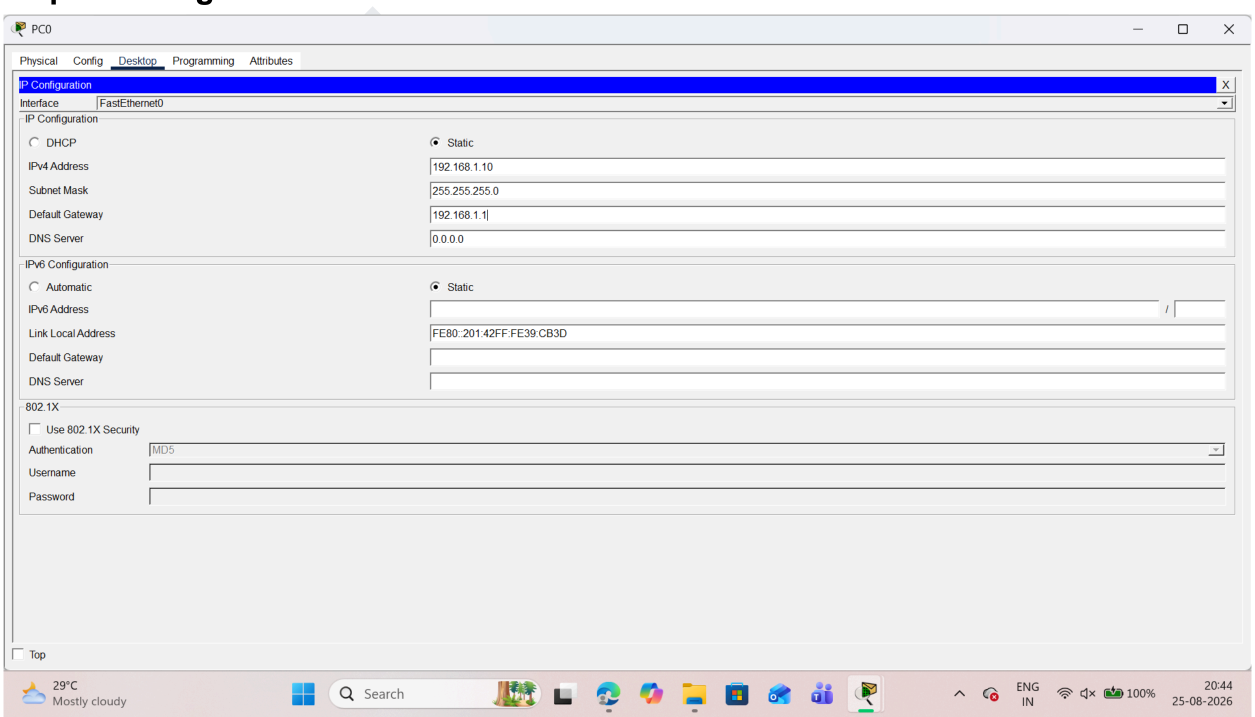
AIM:IP Security (IPsec) Configuration

Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Step 1. Network topology



Step 2. Configure PC IP addresses a.PC0



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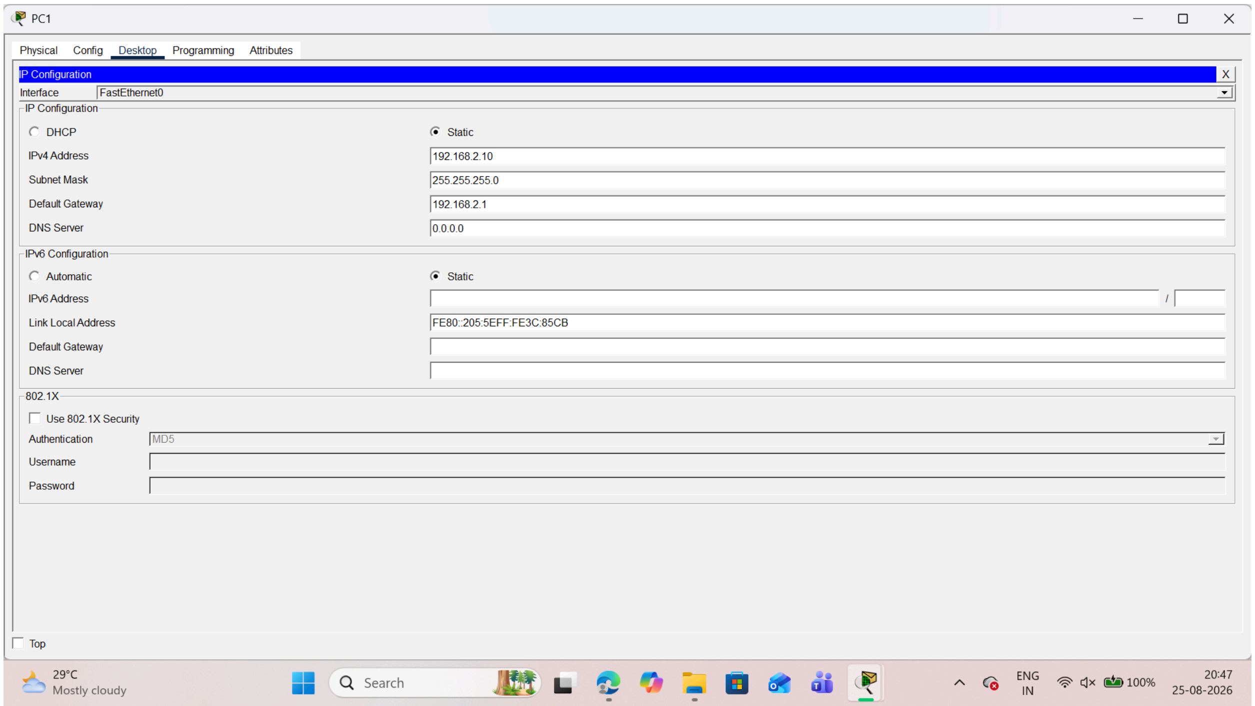
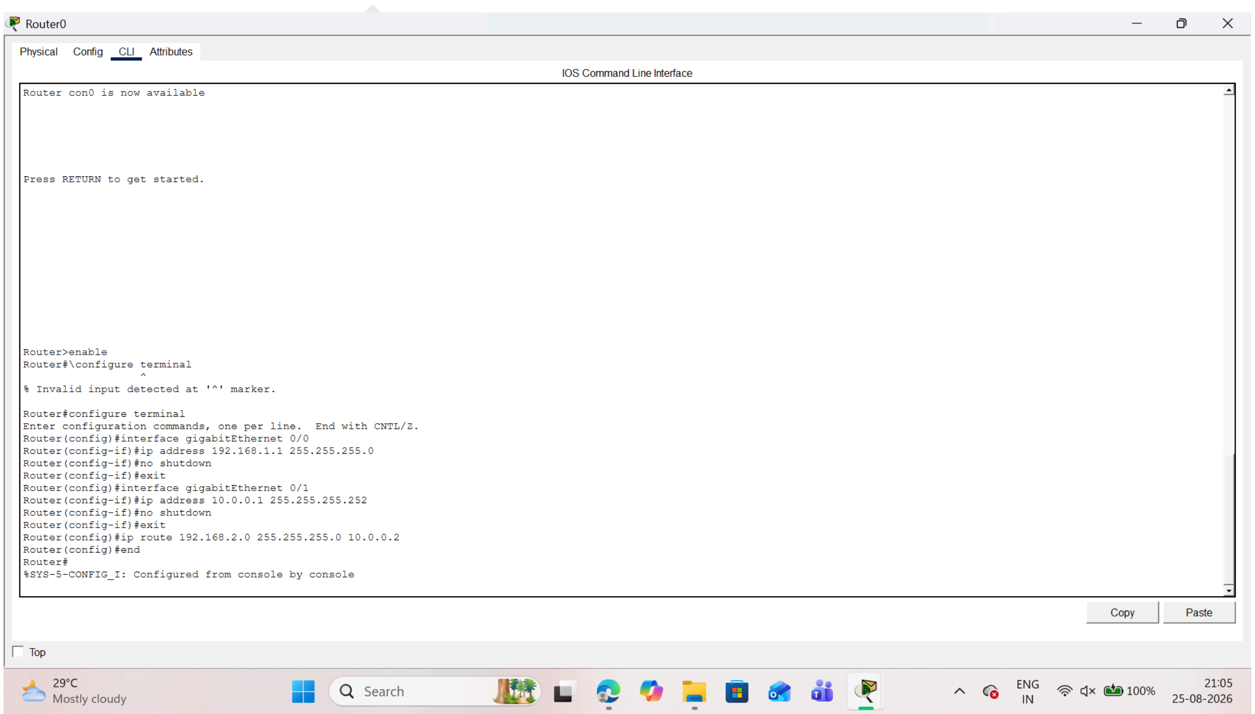
b.PC1

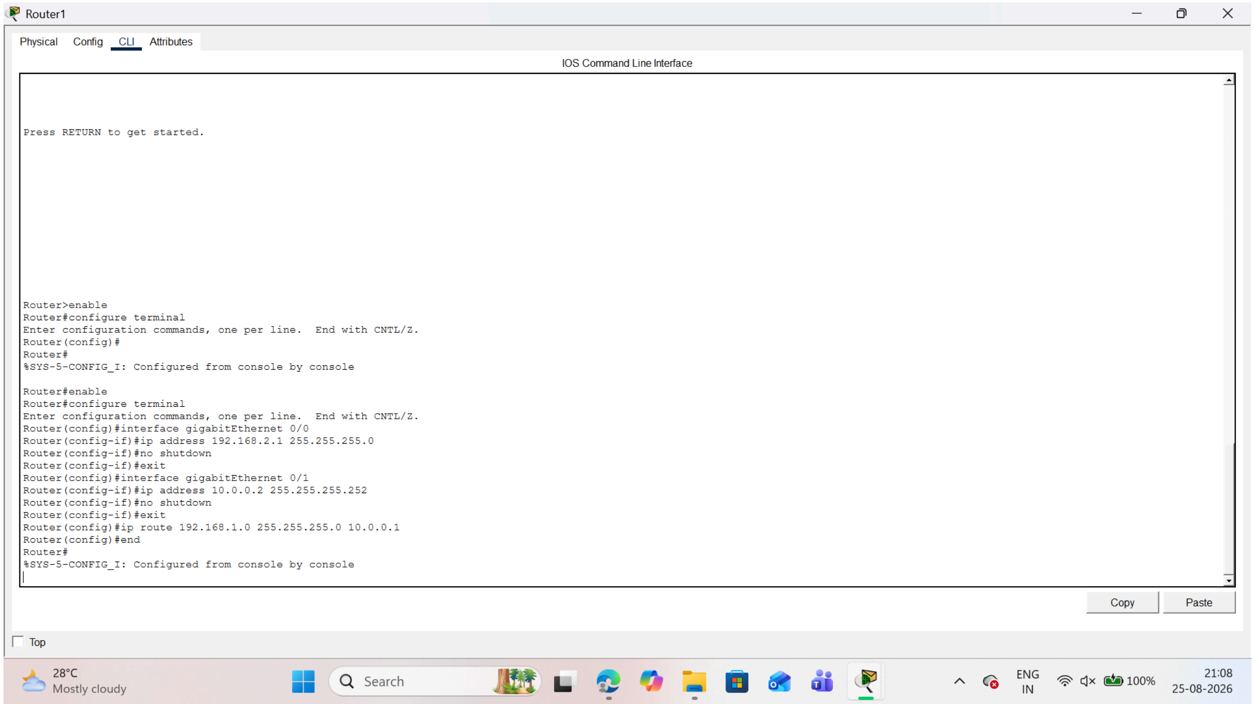


Step 3. Configuring the Routers

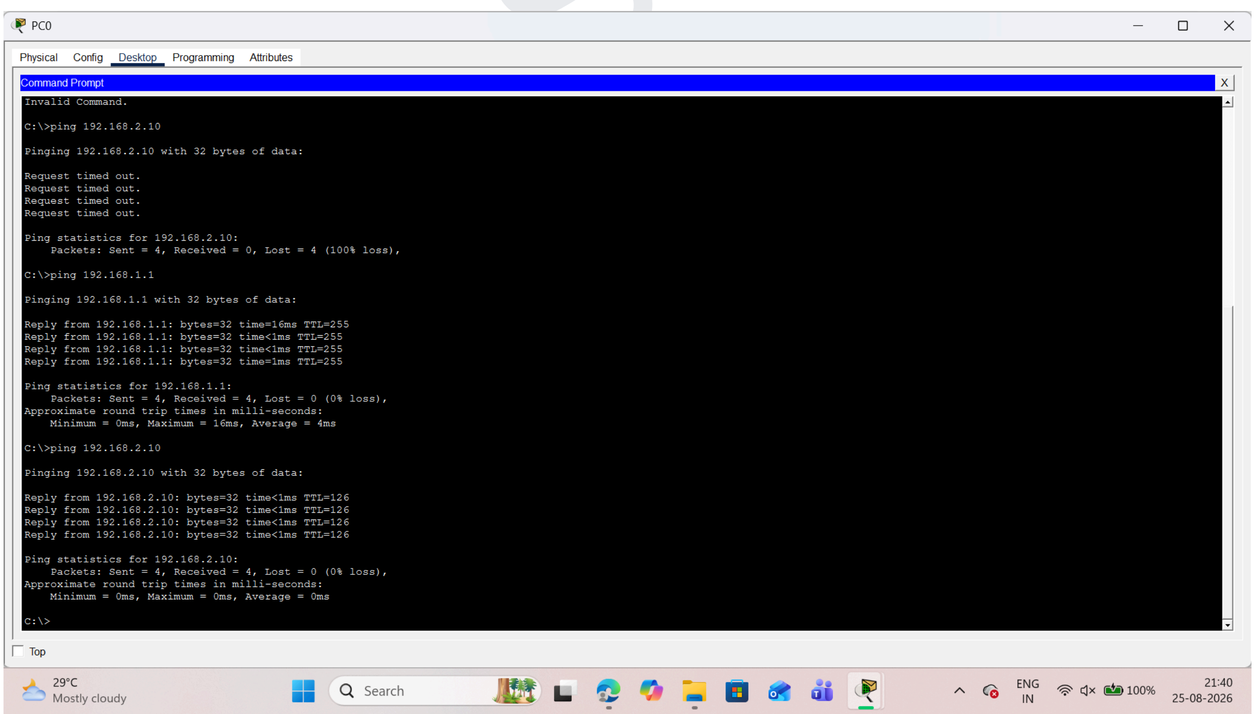
a.Router 0



b.Router 1



Step 4. First test connectivity using ping command



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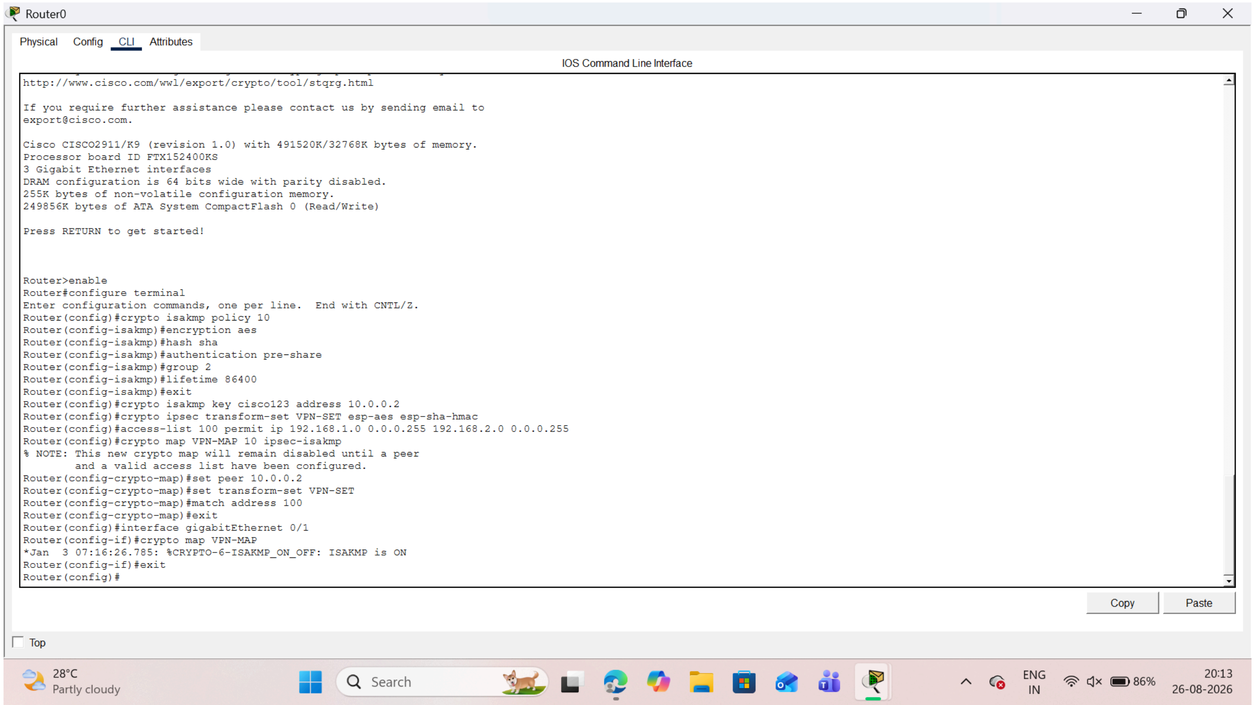
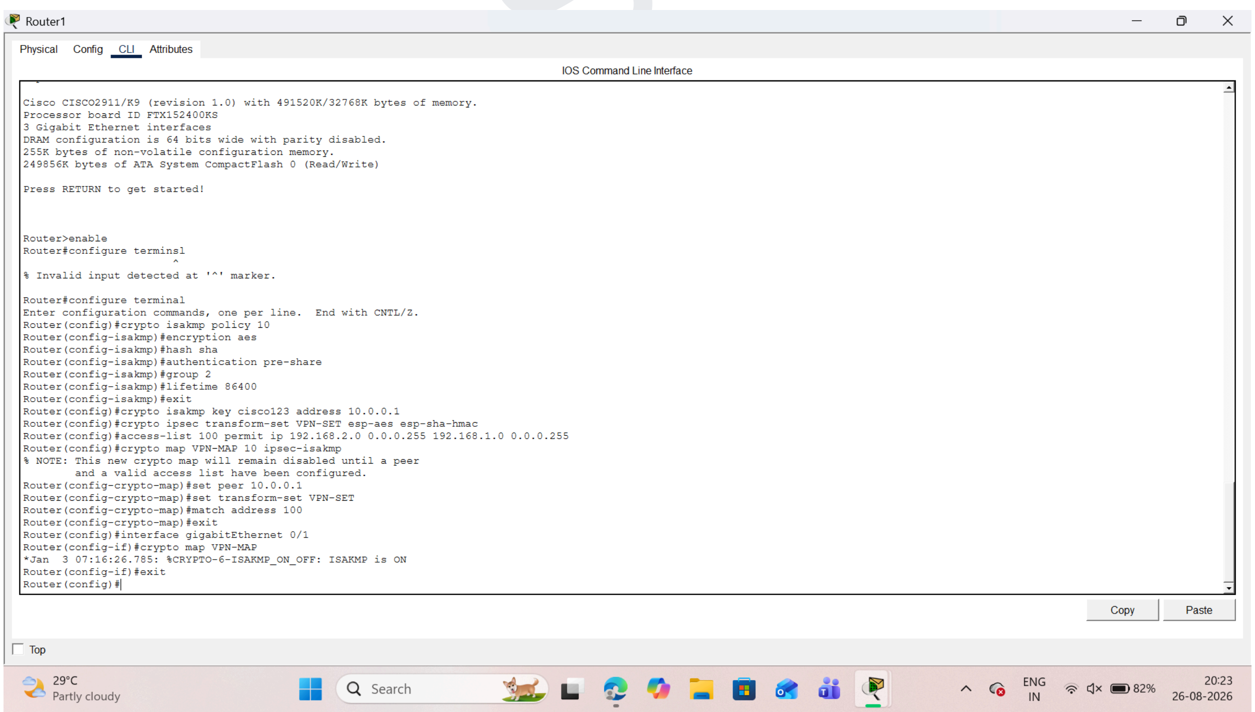
Step 5 . Configure IPsec on Router 0



Step 6 Configure IPsec on Router 1



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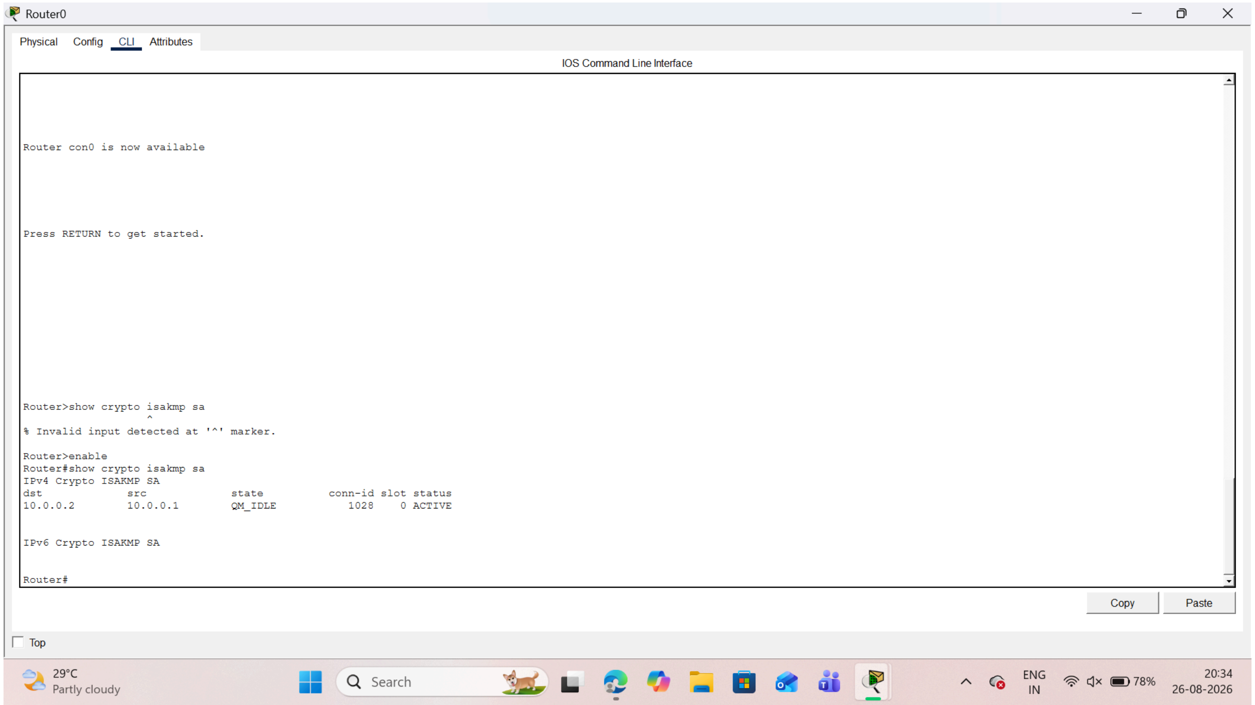
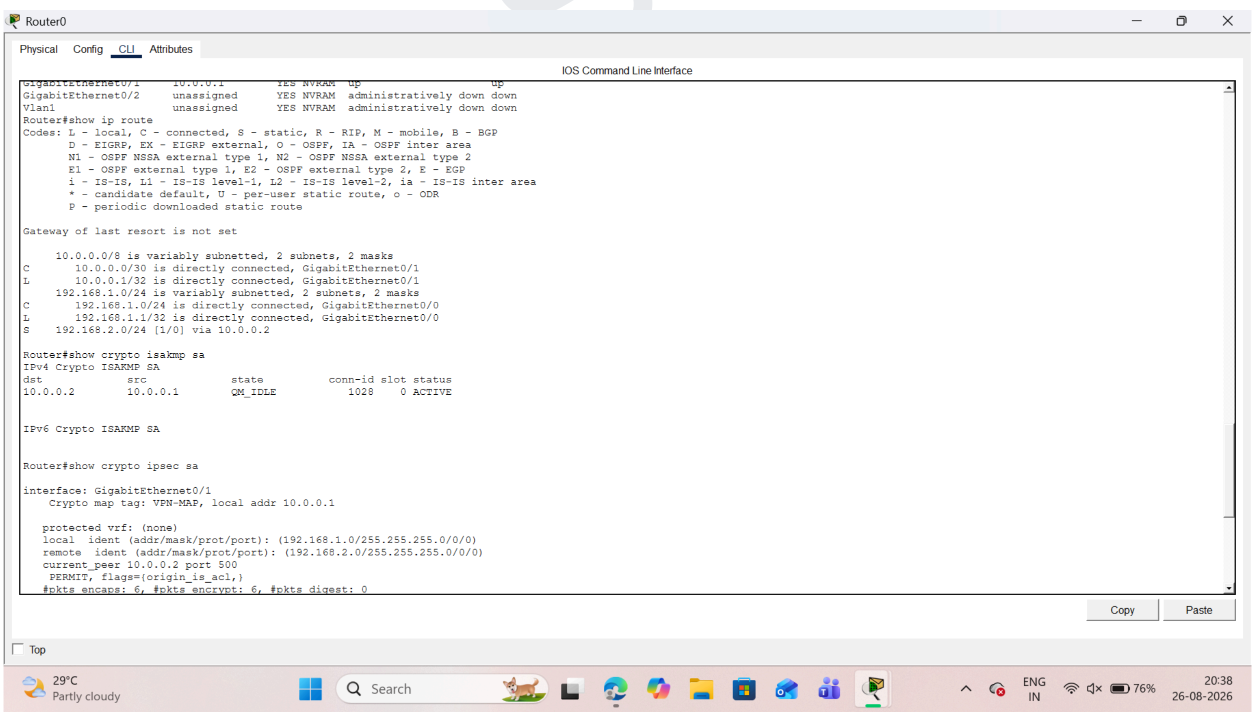
Step 7 Test the IPsec tunnel and verifying





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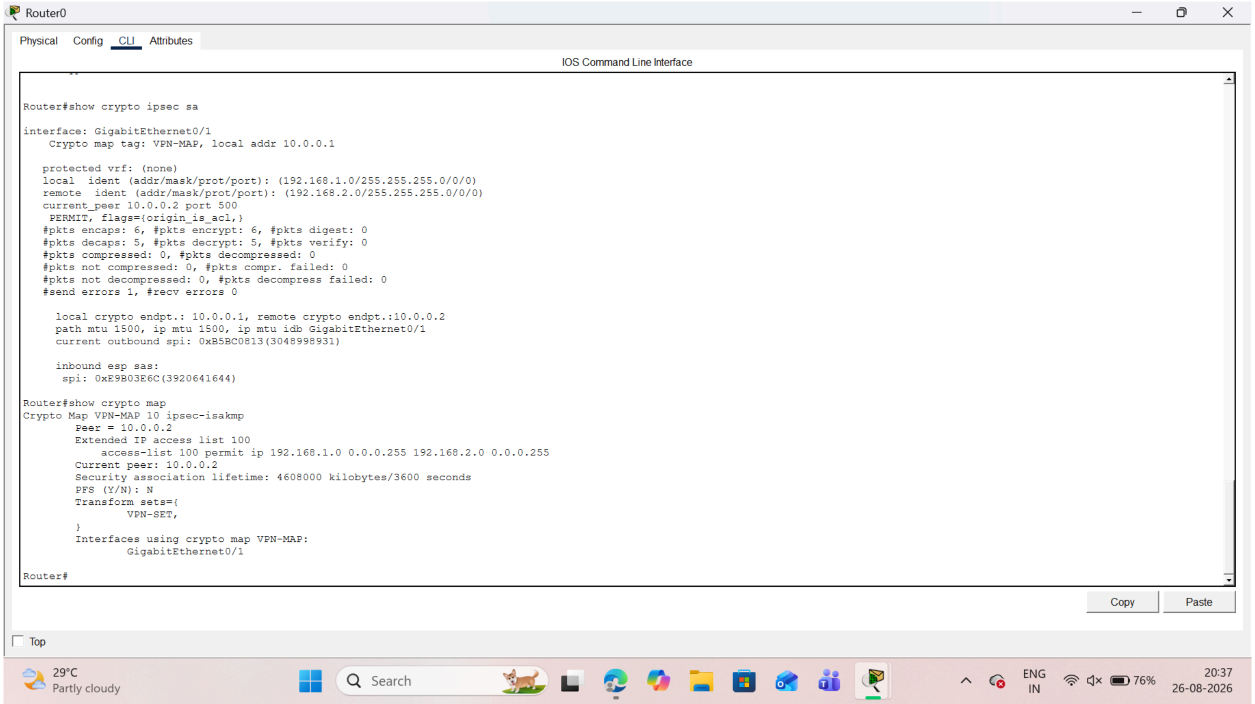
Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.1
protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
#pkts decaps: 5, #pkts decrypt: 5, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0
local crypto endpt.: 10.0.0.1, remote crypto endpt.:10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0xB5BC0813(3048998931)
inbound esp sas:
spi: 0xE9B03E6C(3920641644)
Router#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
Peer = 10.0.0.2
Extended IP access list 100
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Current peer: 10.0.0.2
Security association lifetime: 4608000 kilobytes/3600 seconds
PFS (Y/N): N
Transform sets={
VPN-SET,
}
Interfaces using crypto map VPN-MAP:
GigabitEthernet0/1
Router#
Copy
Paste
Top
29°C
Partly cloudy
Search
ENG
IN
76%
20:37
26-08-2026